# Thesis proposal on Parliamentary Studies

Bocconi University | Department of Social and Political Science

Parliamentary Studies

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**Version:** v1.1 | **Last updated:** April 15, 2026

**Prepared for:** Parliamentary Studies, taught by Prof. Morelli

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April 15, 2026

## Abstract

This article explores the topic of linguistic simplification in parliaments. In 1868 a law broadened the electorate base of UK House of Commons. Can we exploit this setting to measure the effects on linguistic simpilification? Spirling (2016) proved that a broader electoral base has different effects on linguistic simpilification for cabinet and non cabinet members. Building on Spirling (2016), I try to support the causal inference analysis with further investigation of data. In particular, I disentagle the effects of to different measures on the outcome variable provided by Spirling (2016). I will also reduce the time window to empower the effects of observations around 1868 cutoff.

Text as data techinques are used the selct the best method to get a DTM matrix out of the raw speeches. Causal Inference methods are employed on this data. After these steps, I lay down the data generating process, I use a causal diaciclycal graph to build the causal mechanism. These sections should be a suitable theoretical ground for the application of an regression discontinuity design.

# 1 Introducation to the Second Part of the Paper

This second part of the paper is dedicated to test the analysis conducted by Professor Atrhur Spirling (Spirling, 2016). Actually, it provides good reasons to not stop doing research on the topic. In fact, it does not offer any decisive or comprehensive proof of any hypothesis. It suggest that slighlty moving the focus of regressions and inferece closer to the data of 1868 makes us change perpective on linguistic simplification.

*Knowledge of Institutional Setting:* The analysis conducted is not supported by knowledge of institutional setting. Under, this aspect it well reflects what Spirling stated (Spirling, 2016) and does not go deeper. A broader review of the literature on hystorical context in UK and UK parliament is set to be part of future researches that will expand this research proposal.

*Knowledge of Language:* A second limit of the present proposal is linguistic knowledge. In order to strenghthen the results of causal inference it is good to have deep understanding of the language used at the time. This would also shape the selection of n-grams, discriding irrelevant term and valuing relevant expressions.

*Data:* From the point of view of data, I build on the databese provided by Professor Spirling (Spirling, 2015, 2016). I rely on the text as data analysis Professor Spirling conducted in 2015. However, I plan to add topic modelling as a final step. In this way, it will be possible to anlayse correlation between linguistic simplification and extensiono of electorate for each topic identified. In the next section I will provide more details on data emplyed in the study.

*DGP:* It is necessary to understand the data generating process of the topic model in detail in order to match it with the data generating process of speeches. All of the many models reviewed belong to state-of-the art techniques. And all of them have been used in previous political science literature. Therefore, under any likelihood, the literature review will serve as foundation for the extension of the topic modelling analysis.

*Causal Inference:* In this section I present the intuition that challenges the results of Spirling (2016). According to Professor Spirling analysis, after the broadening of electorate base of UK's House of Commons the language used in parliamentary speeches suffered of simplification in terms of lenghth of sentences and words used. Nonetheless, data used in the emprirical tests included observations vary far form the year of the shock (1868). Narrowing down the window of analysis, and robust implying causal inference methodology, I challenge this result and provide a good reason for further testing the main hypothesis.

*Future Research and Limitations:* I conclude this research proposal with a brief list of limitations which are intrinsic of the methodology employed. I complete the paragraph some instances of possible future research that maybe conducted in this topic.

## 2 Causal Inference

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### 2.1 RDD

The identification strategy emplyed in this proposal is regression discontinuity design. Among the classical literature on RDD, I like highlighting two great contributions: Fujiwara (2015) and Gulzar and Pasquale (2017). A more recent study on RDD methodology is presented by Cattaneo et al. (2024).

The equation used is the following:

$$FRE = \beta_0 + \beta_1 \cdot Cutoff + \gamma X + \varepsilon$$

$$FRE = \beta_0 + \beta_1 \cdot C + \gamma X + \varepsilon$$

Identification strategy relies on:

- $C$ is the treatment. It is an exgoneous variable independet from the others called cutoff. In our regression the cutoff is 1868, namely the year in which the reform was introduced
- $Y$ is any of the measures of linguistic simplification employed.

*Controls:* The vector of controls is $X$, while $\gamma$ is the vector of coefficients. It represents the effects of control variables on our outcome variable. The selected controls are:

- Competitiveness (of electoral distric)
- Party

These variables are the same as Spirling (2016).

### 2.2 DinD Approach

Spirling (2016) uses another approach, Difference in Difference, to test the effects electorate reform on linguistic simplification. This design builds on RDD. However, to safely apply it we require another assumption, *parallel trends assumption.* Parallel Trends Assumption states that absent the treatment, control group and treated group would have

had parallel trends. However, It is impossible to test it empirically. Therefore, it requires qualitative support. The equation of DinD is:

$$FRE = \beta_0 + \beta_1 \cdot Cutoff + \beta_2 \cdot Treatment + \gamma X + \varepsilon$$

$$FRE = \beta_0 + \beta_1 \cdot C + \beta_2 \cdot Cabinet + \gamma X + \varepsilon$$

In this setting cabinet status is a dummy variable that represents the treatment.

Identification strategy adopted by Spirling (2016) relies on the same variables as before:

- $C$ is the treatment. It is an exgoneous variable independet from the others called cutoff. In our regression the cutoff is 1868, namely the year in which the reform was introduced
- $Y$ is any of the measures of linguistic simplification employed.

*Controls:* The vector of controls is $X$, while $\gamma$ is the vector of coefficients. It represents the effects of control variables on our outcome variable. The selected controls are:

- Competitiveness (of electoral distric)
- Party

These variables are the same as Spirling (2016).

## 2.3 DAG

Steiner et al. (2017) propose a causal dyciclical graph for RDDs. This is the typical design for this causal inference technique, and it is the design emplyed in this proposal. In particular:

- $U$ stands for unobservable variable.
- $X$ represents speeches made in Parliament by a member
- $D$ is the treatment, thus beign or not beign a cabinet member in a session of the Parliament.
- $c_0$ is an exgoneous variable independet from the others called cutoff.
- $Y$ is any measure of linguistic simplification.

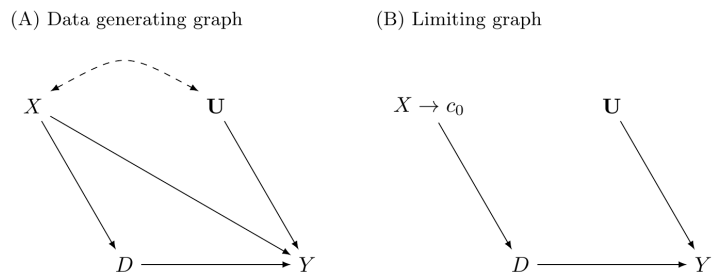


**Figura 1:** *DAG for Regression Discontinuity Design Steiner et al. (2017)*

The DAG formalises the causal mechanism causal inference relies on. It is a usaful way to identify possible weakness of the identification strategy. It strenghthen the analysis showing how the effects of treatment on outcome variable is isolated from the effects of confounders.

In this setting, variables like "Party" and "Competitivness" are confounders. They provide endogenous sources of variation. We take solve this problem controlling for these variables in our regression.

Omitted variable bias happens when we do not control for all confounders. However, it is impossible to eliminate all sources of endogenous variation because some confounders are not observed. In this instance, the solution is not empirical but qualitative. Therefore, it depends on the knowledge of the context under study.

## 2.4 RDD Results

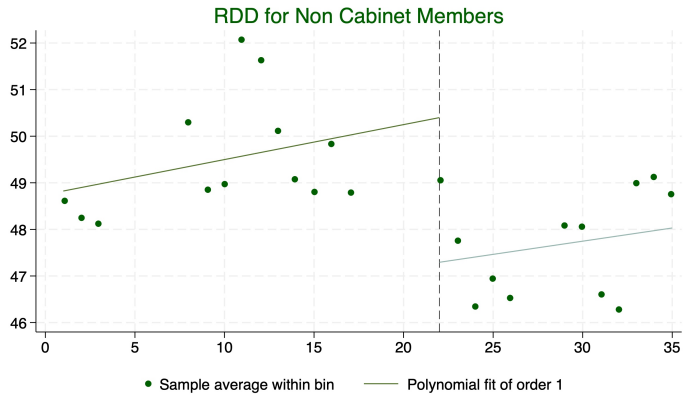


**Figura 2:** *LATE Estimate of RDD for Non Cabinet Members*

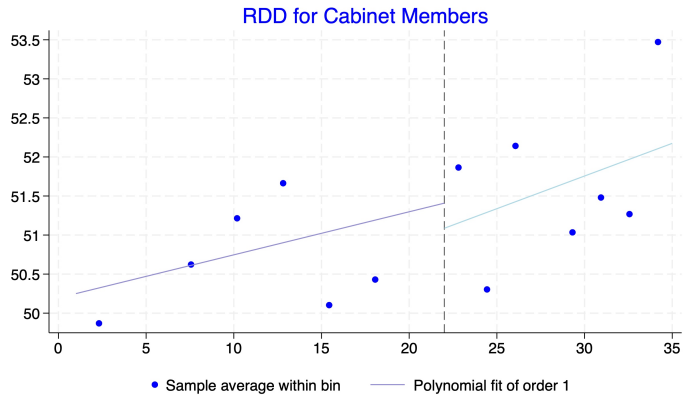


**Figura 3:** *LATE Estimate of RDD for Cabinet Members*

## 2.5 Robustness

### 2.5.1 Bandwidth

Process of selection of optimal window of observation has been extensively studied. Nowadays, most statistical softwares provide command that run optimisation process to automatically select the window.

G. W. Imbens and Angrist (1994) discourges the use of high-order polynomials in the to estame the LATE, they argue that "higher-order polynomials can lead to overfitting and have been found to introduce bias". In the previous pages () we already highlighted the role of overfitting. In particular, we highlighted the contributionof Cranmer and Desmarais (2017).

A way of overcoming the issue of overfitting is using nonparametric kernels (Cunningham, 2021, chap.∽ 6.2.3). However, in this method the selection of bandwidth is crucial. In fact, it "is sensitive to the choice of bandwidth, but more recent work allows the researcher to estimate optimal bandwidths Calonico et al. (2014) and G. Imbens and Kalyanaraman (2012)."

Another issue, namely estimation function, is not stritly related to bandwidth selection. However, Lee and Lemieux (2010) stress the importance of this step. They argue that "allowing different functions on both sides of the discontinuity should be the main results in an RDD paper".

### 2.5.2 Manipulation Tests

In order to strengthen the support to results, I conducted a McCrary Density Test. This test was introduced by McCrary (2008). This test reduces the probability that treated unites "manipulat" the cutoff. In this case it makes us sure of the fact that there is no anticipation effect from Parliament Members. Parliament Members of a party could have decided to rintroduce the reform only when they became sure of having acquired the majority of new voters. This would violate the assumptions that lay at the basics of the regression discontinuity design. The test, conducted for both cabinet and non cabinet, shows that *there is no manipulation.*

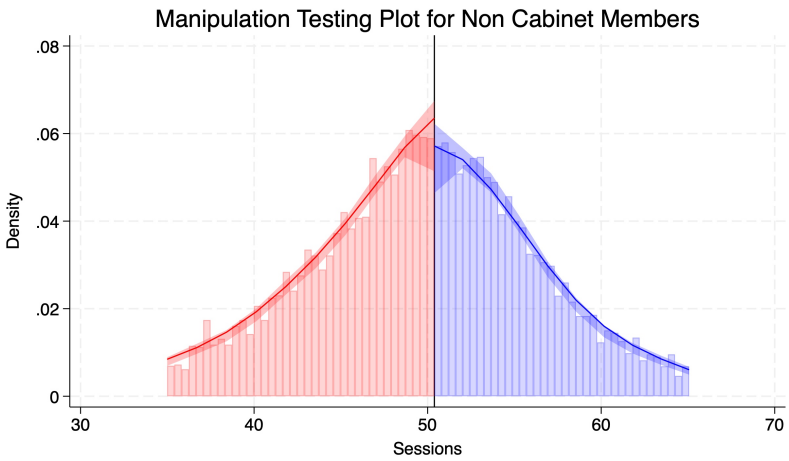


**Figura 4:** *McCrary Manipulation Test for Distribution of Non Cabinet Members around catoff off 1868 election*

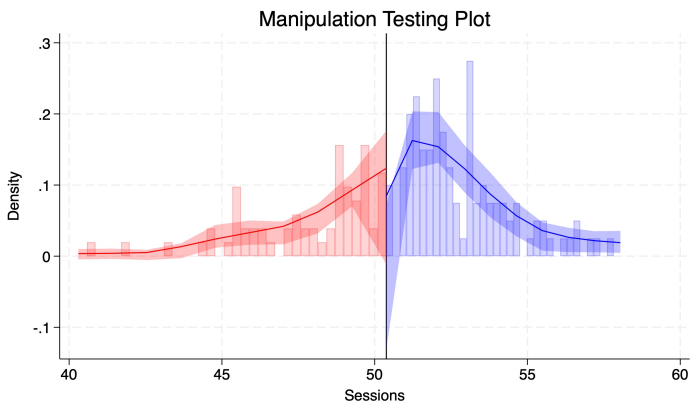


**Figura 5:** *McCrary Manipulation Test for Distribution of Cabinet Members around catoff off 1868 election*

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